

Same game. Different words. Different behavior.

Audited pilots of strategic safety, context, and SAE steering

EXPLORATORY PILOT

Qwen2.5-7B-Instruct F16 - two H100 lanes



The strongest result is a validity warning.

EVIDENCE IN ONE SLIDE

SURFACE

8.4–89.2%

Unsafe-rate span across 18 prompt variants

CONTEXT

+34.0 pp

largest T=0 live effect vs abstract

COMPREHENSION

FAILED

12.5% state update; 17.2% terminal score

ROBUST OUTPUT DIFFERENCES $\neq$ VERIFIED UTILITY OPTIMIZATION

Controlled audit: one exact Qwen checkpoint. Breadth check: five pilot checkpoints under provider-specific protocols. No pooled family effect.

One canonical game, exact state updates.

ENVIRONMENT CONTRACT

VALIDATED ARTIFACT



Safe
progress +1.0



Unsafe
progress +1.5; risk accumulates

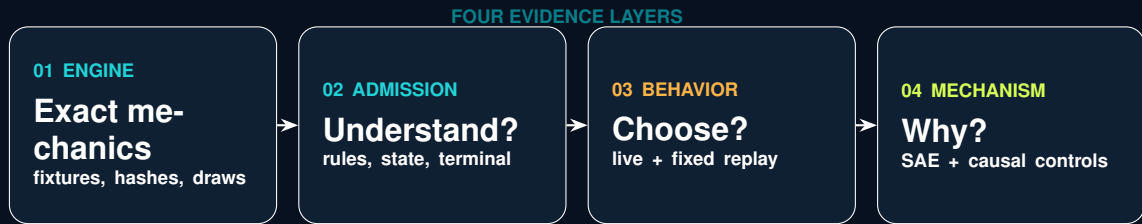
STAGE PAYOFF

$$\pi_i = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}, \quad q_i = p_{\text{risk}}^{\max} \frac{n_i^U}{T}$$

- ▶ simultaneous hidden actions; environment does arithmetic
- ▶ minimum 5 rounds, then 20% stop probability
- ▶ 100-unit prize; winner/tie-specific terminal setback
- ▶ risk caps: 10%, 60%, 90%

Adapted from Fernandez Domingos and Han (2026), arXiv:2607.26034. Human findings are not relabelled as LLM evidence.

Behavior is only the top layer of the audit.

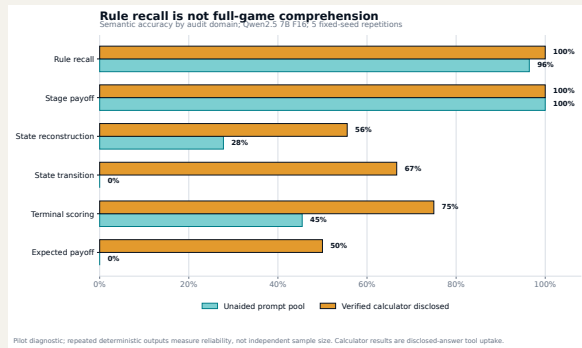


EACH LAYER HAS ITS OWN PROMOTION GATE

Rule recall did not transfer to game arithmetic.

GAME-UNDERSTANDING AUDIT

EXPLORATORY PILOT



RAW PROBES

685

41 items × prompt conditions

SEMANTIC

59.1%

overall accuracy

UNAIDED

52.1%

calculator removed

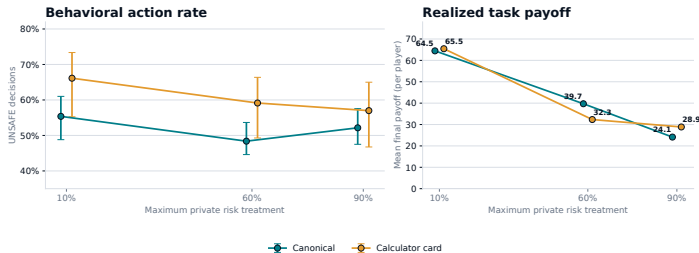
Exact-repeat stability is narrow reliability, not an internal world model. Direct expected-payoff accuracy was 0%.

A calculator changed play; it did not prove understanding.

BEHAVIORAL ABLATION

A correct calculator changes context, not necessarily safety

Paired pilot: 10 races per risk x condition; identical hidden horizons; cluster-bootstrap intervals by repetition



Behavioral pilot only. The decision card enumerates current-round arithmetic but does not predict the opponent or terminal round.

DIAGNOSTIC ONLY

CANONICAL

52.0%

Unsafe; 558 decisions

CALCULATOR

60.8%

Unsafe; 558 decisions

CHANGE

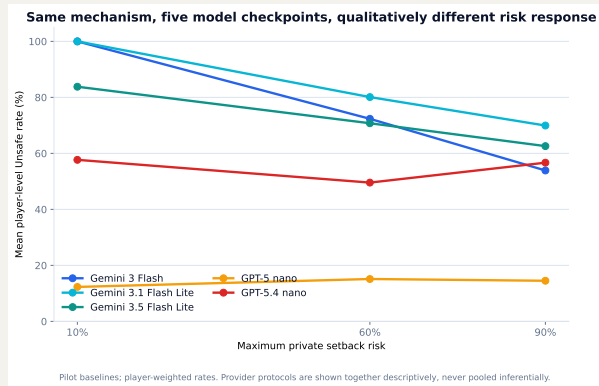
+8.8 pp

30 paired horizon cells

Calculator outputs demonstrate access to disclosed arithmetic. They do not identify internal causal understanding.

Five checkpoints did not share one risk-response curve.

CROSS-CHECKPOINT BASELINE REPLICATION



EXPLORATORY PILOT

BASELINE

150 races

2,790 decisions

CELL SIZE

10 races

per checkpoint $\times$ risk

INFERENCE

SEPARATE

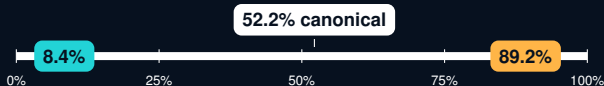
no provider pooling

GPT-5 nano is low-flat; GPT-5.4 nano is non-monotone; three Gemini checkpoints decline with risk. This is qualitative heterogeneity, not a vendor ranking.

Prompt surface moved the same model across 81 points.

18-VARIANT SENSITIVITY GRID

EXPLORATORY PILOT



Extreme whole-trajectory variants

action order reversed: 8.4% Unsafe

risk text near response: 89.2% Unsafe

COVERAGE

540 races

10,044 dependent decisions

PARSE FAILURES

0

all output contracts valid

FIRST ROUND

83.3% flips

emotional framing vs canonical

Whole-trajectory differences include endogenous feedback. First-round flips compare matched initial states and seeds.

Eight stories preserve one mathematical game.

CONTEXT-SKIN INVARIANCE DESIGN

VALIDATED ARTIFACT

CONTROL SKINS

- ▶ technology race
- ▶ abstract contest

REALISTIC / FICTIONAL PAIRS

- ▶ logistics contract / crystal guild
- ▶ hospital deployment / colony life support
- ▶ robotic expedition / fictional cartography

Payoff-preserving relabeling



Safe=P and Safe=Q crossed
same state, mechanics, model digest

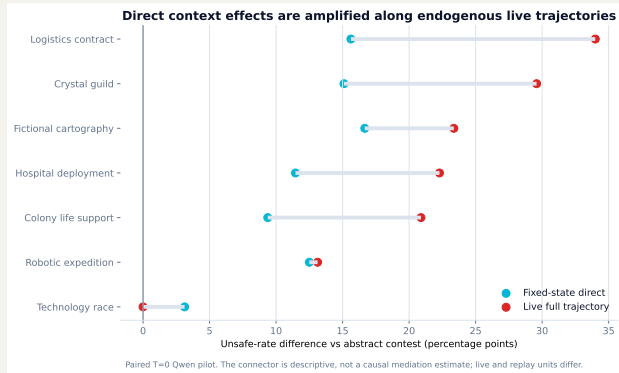
NO SAFE / UNSAFE LABEL LEAK

Primary: 768 live races at temperature 0; 96 engine-reachable states $\times$ 8 skins $\times$ 2 mappings at temperature 0.

Direct context response grows along live trajectories.

FIXED-STATE REPLAY VERSUS ENDOGENOUS PLAY

DIAGNOSTIC ONLY



ENTRY

1,344 /
1,344

paired actions agree

LIVE MAX

+34.0 pp

logistics vs abstract

DESCRIPTIVE GAP

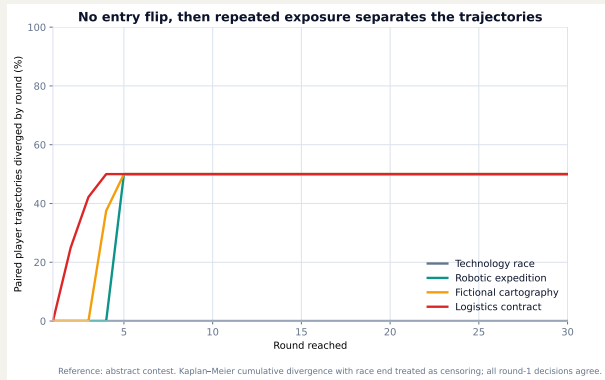
+18.4 pp

live minus fixed, max

Both protocols use T=0. Live and replay units differ: the connector is consistent with repeated exposure and feedback amplification, but is not a causal mediation estimate.

No entry flip did not mean robust trajectories.

ROUND-BY-ROUND DIVERGENCE



DIAGNOSTIC ONLY

ROUND 1

0 flips

all paired contexts

FIRST SPLIT

2.5–5

conditional median round

CONTROL

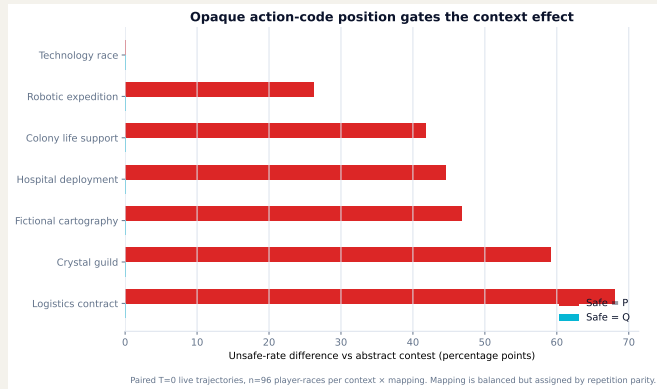
0

technology divergences

Kaplan–Meier cumulative divergence treats race termination as censoring. Entry-only robustness checks would miss the state-conditional separation.

Opaque labels exposed a dominant code-position shortcut.

P/Q MAPPING DIAGNOSTIC



DIAGNOSTIC ONLY

SAFE=P

6 / 7

contexts diverge

SAFE=Q

0 / 7

contexts diverge

MAX SHIFT

+68.0 pp

logistics, Safe=P

Mapping was balanced but assigned by repetition parity. A frozen 1,536-race diagnostic now crosses both mappings inside every seed block; launch awaits current GPU NodePorts.

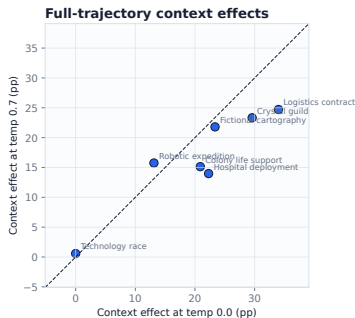
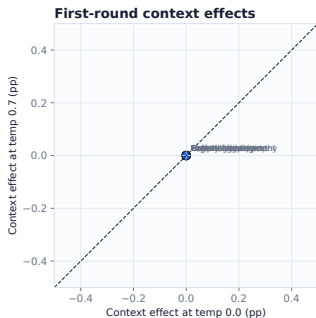
Temperature changes trajectories more than first moves.

DECODING ROBUSTNESS: T=0 VERSUS T=0.7

DIAGNOSTIC ONLY

Context-effect rank and sign stability

Each effect is context minus abstract control within the same temperature and CRN race key.



OVERALL

+3.04 pp

95% CI +2.19 to +3.97

ACTIONS

88.2%

paired agreement

TRAJECTORIES

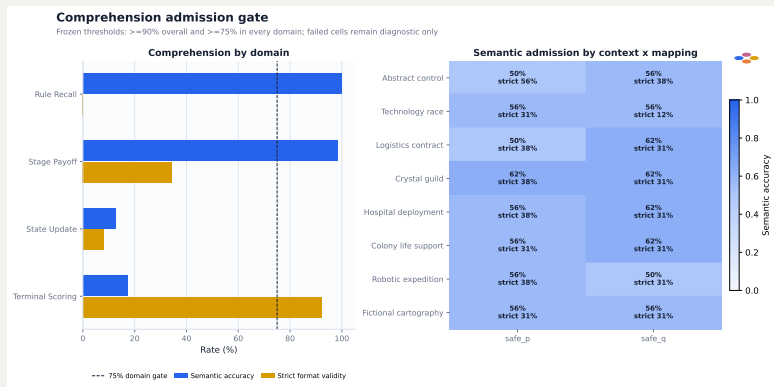
62.6%

exact player paths

Full-effect rank $\rho = .857$; first-round delta 0.00 pp. Mechanism/template hashes and CRN horizons match, but whole staged-source hashes differ. No comprehension claim.

The context behavior study failed its admission gate.

COMPREHENSION BEFORE INTERPRETATION



DIAGNOSTIC ONLY

RULE RECALL
100%

semantic accuracy

STATE UPDATE
12.5%

semantic accuracy

TERMINAL
17.2%

semantic accuracy

Interpretation: prompt-conditioned action production under identical mechanics - not verified informed optimization.

Recognition audit v1 failed; v2 repaired the contract.

PROTOCOL DEBUGGING IS A RESULT

V1 - REJECTED

0 / 320 admitted

Contract required a named candidate even for generic resemblance. All 960 attempts therefore failed parsing. Raw output was preserved, not rescored.

V2 - ACCEPTED

16 / 16 valid

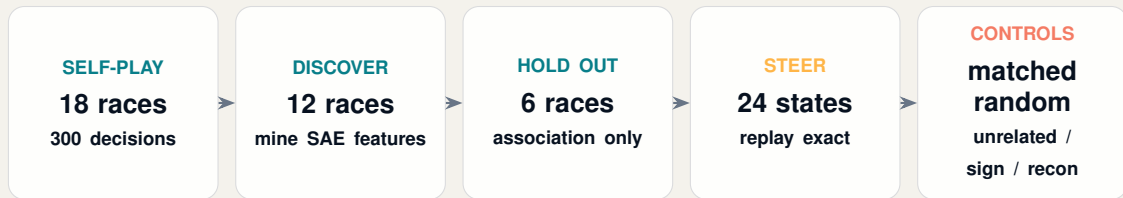
All responses: generic structural resemblance, high confidence, no named game. Mapping agreement held for all 8 skins.

Self-report neither proves nor rules out memorization, contamination, recognition, or latent understanding.

AUC answers “decodable?” - steering asks “causal?”

ACTUAL-SELF-PLAY FAST-SAE DESIGN

VALIDATED ARTIFACT



Exact full-sequence likelihood: ACTION: SAFE vs ACTION: UNSAFE; intervention at layer 12, final prompt token.

Native Qwen2.5-7B-Instruct revision and FAST-SAE revision are pinned in the manifest.

Three SAE features predicted held-out actions.

ASSOCIATION STAGE

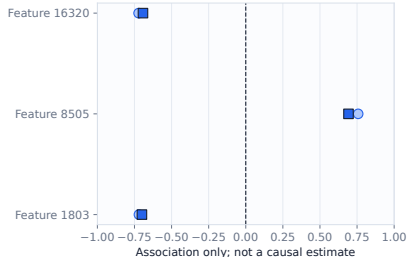
DIAGNOSTIC ONLY

Selected FAST-SAE feature associations

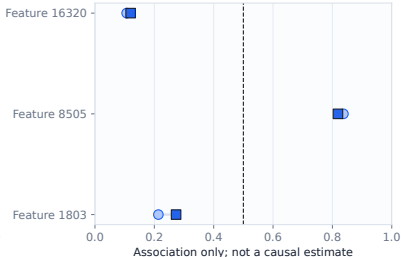
Layer 12, final prompt token; AUC direction is not re-oriented, so values below 0.5 denote inverse coding.

○ Discovery (12 races) ■ Held-out eval (6 races)

Pearson correlation



Unsafe-action AUC



F8505

0.819

held-out action AUC

F16320

0.120

inverse-coded AUC

F1803

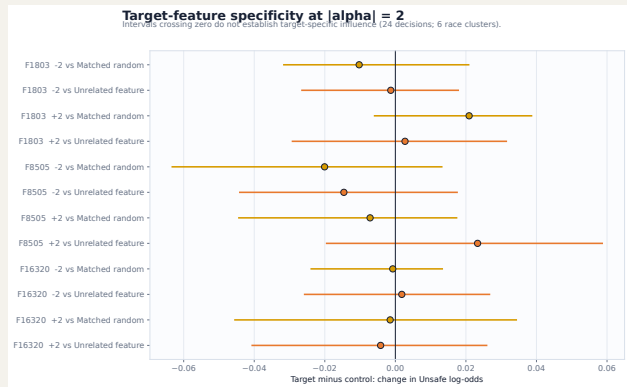
0.273

inverse-coded AUC

Held-out absolute correlations were 0.692–0.700. Association is useful for screening; it is not a neuron-level reason.

Target steering did not beat its controls.

FIXED-STATE CAUSAL AUDIT



DIAGNOSTIC ONLY

CONTRASTS

0 / 12

CIs exclude zero

RECON FLIPS

12.5%

SAE reconstruction not neutral

EVAL CLUSTERS

6

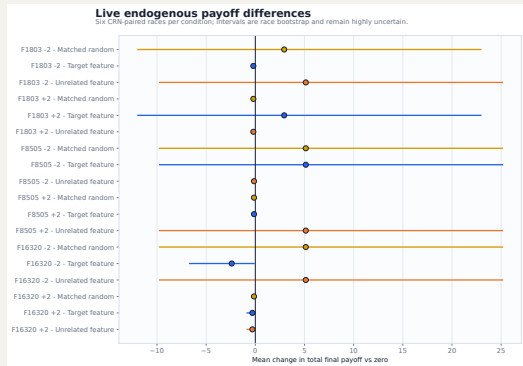
race-limited uncertainty

No feature-specific causal-control claim. Dose curves were neither consistently monotone nor sign-reversing.

Live steering changes trajectories, then feedback takes over.

ENDOGENOUS SELF-PLAY EFFECTS

DIAGNOSTIC ONLY



- ▶ 114 live trajectories over 6 common-random-number seeds
- ▶ direct comparable flips: 1.1–3.9%
- ▶ target/control action sequences matched 68.1% on average
- ▶ payoff deltas sit on the same scale as controls

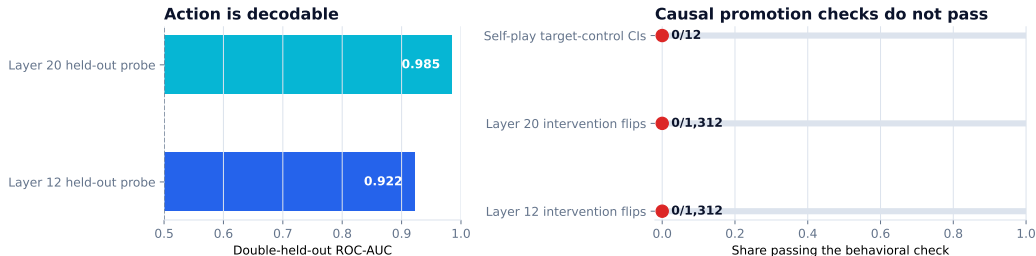
After the first action divergence, changed history and state make later differences total trajectory effects, not fixed-state causal effects.

Context is decodable at two layers - causality still fails.

CONTEXT FAST-SAE SMOKE

DIAGNOSTIC ONLY

Decodable representation without demonstrated feature-specific control



Pinned Qwen2.5-7B-Instruct + FAST-SAE. AUC is association; intervention flips and target-control intervals test behavioral specificity.

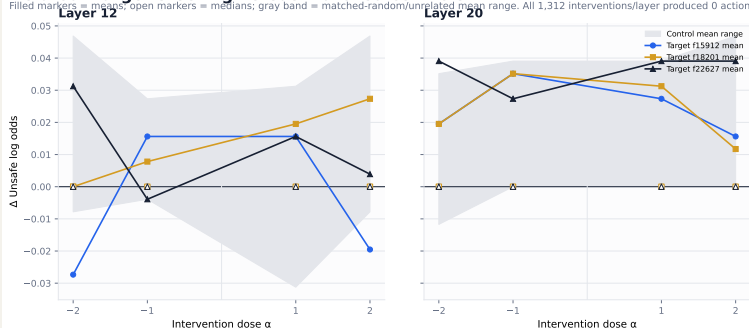
Double holdout crosses unseen state trajectories and unseen context pairs ($n = 56$). High AUC measures representational availability; zero held-out action flips and zero target-control intervals passing preserve the causal boundary.

Layer 20 wins the screen, not the causal claim.

CONTEXT-SAE PROMOTION DECISION

Held-out target steering versus matched controls

Filled markers = means; open markers = medians; gray band = matched-random/unrelated mean range. All 1,312 interventions/layer produced 0 action flips.



DIAGNOSTIC ONLY

LAYER 20

PROMOTE

capture + analysis

LAYER 12

HOLD

smoke robustness point

STEERING

STOP

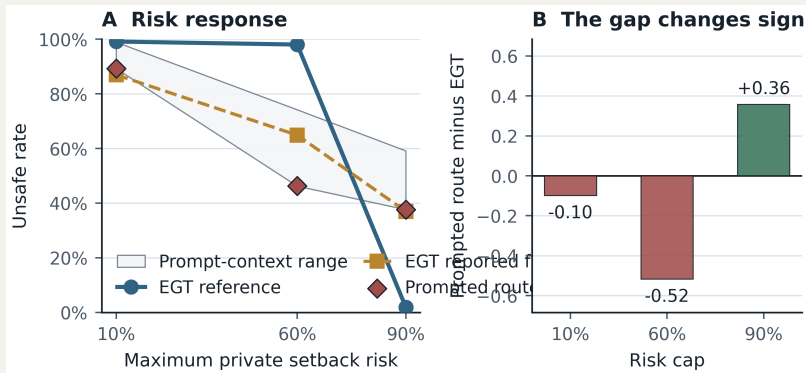
0 held-out flips

Require at least 10 discovery flips before steering, or preregister continuous log-odds feature selection.

Evolution predicts a risk phase change; Qwen does not.

REDUCED EGT RECONSTRUCTION VERSUS LLM SELF-PLAY

VALIDATED ARTIFACT



RISK .1 - AU

99.2%

predicted Unsafe

RISK .6 - CAS

98.0%

predicted Unsafe

RISK .9 - CS

1.9%

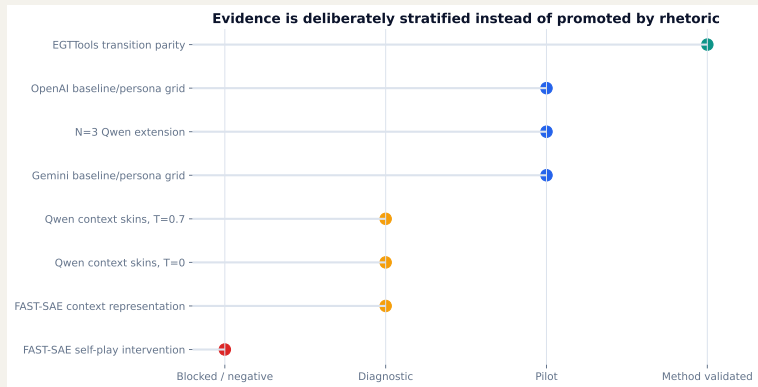
predicted Unsafe

Faithful, not bitwise: author code/seeds are unavailable. Official EGTTools source parity passed (transition max error 1.11×10^{-16}).

LLM rates remain context-dependent and are not evolutionary samples.

Evidence is promoted by gates, not by effect size.

EVIDENCE LADDER



METHOD

VALIDATED

engine + EGT parity

BEHAVIOR

PILOT

breadth, not population

MECHANISM

WITHHELD

controls do not pass

Supported: exact mechanics, localized comprehension failures, and prompt-conditioned behavior. Not supported: informed optimization, stable family effects, or a semantic Unsafe feature.

Promotion requires stronger controls, not more rows alone.

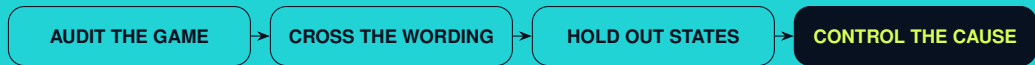
NEXT EVIDENCE GRID

- ▶ launch frozen 1,536-race context $\times$ mapping diagnostic
- ▶ cross both mappings within every identical live race seed
- ▶ at least three model families and fixed decoding
- ▶ replay logged states to separate direct effects from feedback
- ▶ freeze layer, feature, sign, alpha, and controls before evaluation
- ▶ use multiple matched-random directions as an empirical null
- ▶ require neutral reconstruction / zero-hook gates
- ▶ report race-level, seat-level, and strategy-level estimands together

PRIMARY UNIT OF UNCERTAINTY: WHOLE RACE / MATCHED STATE - NOT DECISION ROW

Same payoff matrix. Different context. Different output.

But the model did not pass the gate required to call that behavior informed optimization.



Reproducible artifacts in `results/` • interactive trajectory demo + vector evidence figures

References and reproducibility.

SELECTED SOURCES

VALIDATED ARTIFACT

- ▶ Fernandez Domingos, E. and Han, T. A. (2026). *Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment*. arXiv:2607.26034.
- ▶ *Same Game, Different Story* (2026). Payoff-preserving contextual robustness for LLM game behavior. arXiv:2607.19670.
- ▶ *Framing the Game* (2025). Context effects in game-theoretic LLM evaluation. arXiv:2503.04840.
- ▶ *RouteSAE* (2025). Sparse autoencoders for interpreting mixture-of-experts routing. arXiv:2503.08200.

```
model digest: 59805ce4...ff16c
live context: 768 races / 13,680 decisions
fixed replay: 96 states / 1,536 cells
cross-checkpoint baseline: 150 races / 2,790 decisions
FAST-SAE self-play: 18 races / 300 decisions / 114 steered trajectories
```

Exact configs, raw responses, manifests, hashes, CSV summaries, PNGs, and vector PDFs are stored beside each analysis report.